RAJALAKSHMI ENGINEERING COLLEGE

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MINI PROJECT REPORT

On

Vehicle Number Plate Recognition (ANPR)

Image Processing and Computer Vision Project

Submitted by

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ABSTRACT

Automatic Number Plate Recognition (ANPR) is a critical component of modern intelligent transportation systems, enabling automated vehicle identification for traffic management, security surveillance, and toll collection. This report presents the development and implementation of a robust ANPR pipeline designed to detect and recognize vehicle license plates from diverse image datasets. The proposed system follows a modular architectural approach, consisting of four primary stages: dataset acquisition, image preprocessing, plate localization, and optical character recognition (OCR). The pipeline utilizes a combination of grayscale conversion and Gaussian blurring to reduce noise and enhance edge definition. For localization, a contour-based detection strategy is employed, leveraging Morphological Snake Edge Refinement (MSER) and geometric filtering to isolate the plate region from the background. The final stage employs Tesseract OCR to extract alphanumeric characters, which are subsequently cleaned and normalized to ensure high recognition accuracy. The system was evaluated using the ud-smart-city license plate dataset, focusing on balancing detection sensitivity with precision to avoid over-filtering. The results demonstrate a functional end-to-end pipeline capable of accurately identifying license plates under standard lighting conditions. This report details the technical challenges encountered during the implementation, the design decisions made to optimize the detection algorithm, and the final performance metrics of the combined pipeline.

Keywords: Automatic Number Plate Recognition (ANPR), Contour Detection, Computer Vision, Image Preprocessing, OpenCV, Tesseract OCR.

1. INTRODUCTION

1.1 Overview

Automatic Number Plate Recognition (ANPR) is a specialized application of computer vision and optical character recognition (OCR) that enables the automatic identification of vehicle registration plates. In an era of increasing urbanization and vehicle density, the ability to automate the monitoring of vehicle movement is essential for maintaining public safety, managing urban traffic flow, and implementing automated payment systems such as electronic toll collection.

The core challenge of ANPR lies in the variability of the environment. License plates must be detected and read across a wide array of conditions, including varying lighting, different camera angles, diverse plate designs across different regions, and the presence of environmental noise. To address these challenges, a multi-stage pipeline is required, where each stage progressively refines the image data to isolate the characters of interest.

1.2 Objectives of the Project

The primary objective of this project is to develop a modular and efficient ANPR pipeline that can reliably transition from a raw image to a cleaned alphanumeric string. The specific goals include:
- Implementing a robust image preprocessing stage to normalize input images.
- Developing a plate localization algorithm that can distinguish the license plate from the rest of the vehicle and the background.
- Integrating a high-performance OCR engine to convert the detected plate image into text.
- Building a comprehensive testing framework to evaluate the accuracy and reliability of the system across a standardized dataset.

1.3 Scope and Methodology

This report focuses on the implementation of a contour-based detection system using Python and the OpenCV library. The methodology follows a linear pipeline: starting with dataset loading from the 'ud-smart-city' source, followed by grayscale and blur processing, the application of Morphological Snake Edge Refinement (MSER) for candidate region identification, and finally the use of Tesseract OCR for character extraction.

The development process emphasizes modularity, ensuring that each component (Preprocessing, Detection, and OCR) can be tuned and tested independently before being integrated into the final system. The project specifically targets standard alphanumeric plates, ensuring that the final output is normalized to uppercase text for consistency.

1.4 Report Structure

The remainder of this report is organized as follows:
- Section 2 (Dataset) describes the source and characteristics of the images used for training and testing.
- Section 3 (Preprocessing) details the image enhancement techniques used to prepare the images for detection.
- Section 4 (Algorithm) explains the mathematical and geometric logic used for plate localization and character recognition.
- Section 5 (Results) analyzes the performance of the system and discusses the success rate.
- Section 6 (Conclusion) summarizes the findings and suggests future improvements.
- Section 7 (References) lists the libraries and datasets utilized.

2. DATASET

2.1 Source and Acquisition

The effectiveness of any computer vision pipeline is heavily dependent on the quality and diversity of the data used for validation. For this project, the 'ud-smart-city/license-plate-dataset' was selected as the primary data source, sourced from Hugging Face. The dataset can be accessed at the following URL: https://huggingface.co/datasets/ud-smart-city/license-plate-dataset.

This dataset provides a curated collection of vehicle images specifically designed for license plate detection and recognition tasks.

The dataset consists of approximately 120 images, featuring vehicles in various orientations and environments. These images serve as a critical benchmark for testing the pipeline's ability to handle real-world variability, including different vehicle colors, background clutter, and varying degrees of image resolution.

2.2 Data Characteristics

The images in the dataset are diverse in terms of content, presenting several challenges for the detection algorithm:
- Environmental Noise: Images include various backgrounds, such as urban roads, parking lots, and nature, which can introduce false-positive contours.
- Lighting Conditions: The dataset contains images with different lighting intensities, affecting the contrast between the license plate and the vehicle body.
- Plate Variety: While the focus is on standard alphanumeric plates, the dataset includes variations in plate size and aspect ratio relative to the overall image.

2.3 Dataset Management

To ensure an organized workflow, a structured directory system was implemented:
- raw/: This directory stores the original images downloaded from the source.
- processed/: This directory is used to store images after they have undergone preprocessing steps, allowing for an audit trail of how the image data evolves through the pipeline.

The use of a small, controlled sample size (initially testing on 10 images) allowed for rapid iteration of the detection parameters before scaling to the full dataset of 120 images, preventing the "over-fitting" of parameters to a specific set of images.

2.4 Data Labeling and Ground Truth

The dataset provides the necessary ground truth for evaluating the accuracy of the OCR stage. The expected output for each image is a specific alphanumeric string representing the license plate. This allows for the calculation of precision and recall metrics, comparing the system's predicted text against the actual plate numbers.

3. PREPROCESSING

3.1 Rationale for Preprocessing

The raw images acquired from the dataset contain significant amounts of noise, varying color intensities, and complex backgrounds that can interfere with the plate detection algorithm. The primary goal of the preprocessing stage is to simplify the image data, enhancing the contrast between the license plate and its surroundings while removing irrelevant details. By normalizing the input, we ensure that the subsequent contour detection and OCR stages operate on high-quality, standardized data.

Fig. 1. Raw Image

The original images, stored in the `dataset/raw/` directory, contain full-color information, including the vehicle's paint, environmental lighting, and background details. While visually complete, this level of detail is counterproductive for the detection of geometric shapes like license plates.

3.2 Grayscale Conversion

The first step in the pipeline is the conversion of the RGB image to a single-channel grayscale image. Since the structural information (edges and shapes) is more critical for plate detection than color information, grayscale conversion reduces the dimensionality of the data from three channels (Red, Green, Blue) to one. This significantly reduces the computational overhead and removes color-based noise.

Fig. 2. Grayscaled Image

3.3 Gaussian Blurring

Even after grayscale conversion, images often contain high-frequency noise—small, sharp variations in intensity that can be mistaken for edges by the Canny or MSER algorithms. To mitigate this, a Gaussian Blur is applied. This technique involves convolving the image with a Gaussian kernel, which effectively "smooths" the image by averaging local pixel values. This process suppresses fine-grain noise and softens unimportant textures, ensuring that only the most prominent edges are detected.

Fig. 3. Blurred Image

3.4 Result of Preprocessing

The combined effect of these operations transforms the raw input into a format optimized for localization. The resulting images, saved in the `dataset/processed/` directory, exhibit reduced noise and enhanced structural clarity.

Fig. 4. Processed Image

The processed image demonstrates the transition from a complex, color-rich raw image to a simplified, noise-reduced grayscale version. This transformation increases the signal-to-noise ratio, allowing the Morphological Snake Edge Refinement (MSER) algorithm to identify candidate regions based on actual structural boundaries rather than random pixel fluctuations.

4. ALGORITHM

4.1 Plate Detection Strategy

The core of the ANPR system is the localization of the license plate within the larger image. This is achieved through a multi-stage geometric and morphological analysis. Rather than relying on a single strict filter, the system employs a "permissive-to-strict" cascade filtering approach to ensure that potential candidates are not discarded prematurely while maintaining a high precision rate.

4.2 Morphological Snake Edge Refinement (MSER)

The primary tool for region proposal is Morphological Snake Edge Refinement (MSER). MSER is particularly effective for
ANPR because it identifies regions that maintain a constant intensity over a range of thresholds, which is a characteristic property of the high-contrast text on a license plate.

The algorithm scans the image to find "stable" regions (blobs) that correspond to characters or the plate itself. These regions are then grouped to identify larger rectangular structures. This approach is significantly more robust than simple Canny edge detection, as it is less susceptible to noise and varying lighting conditions.

4.3 Contour Analysis and Geometric Filtering

Once MSER identifies candidate regions, the system applies a series of geometric constraints to filter out non-plate contours. A valid license plate typically adheres to specific physical proportions. The following criteria are used to validate candidates:

- Aspect Ratio: License plates have a characteristic width-to-height ratio. Any detected contour that falls outside a defined range (typically 2:1 to 5:1) is discarded.
- Solidity: This measures the ratio of the contour area to the area of its minimum bounding rectangle. A high solidity indicates a dense, rectangular shape, which is typical of a license plate.
- Area Constraints: To avoid detecting tiny noise artifacts or massive background blocks, a minimum and maximum area
threshold is applied.

Fig. 5. Detected Plate

The detected plate image shows the result of the localization process, where a bounding box has been accurately placed
around the license plate, isolating it from the vehicle body.

4.4 Optical Character Recognition (OCR)

Once the plate region is isolated, it is cropped and passed to the Tesseract OCR engine. Tesseract performs a sequence of operations to convert the image pixels into text:

1. Binarization: The cropped plate image is converted to a binary (black and white) format to maximize the contrast between the characters and the plate background.
2. Character Segmentation: The algorithm identifies individual characters based on whitespace and connectivity.
3. Pattern Matching: Tesseract compares the segmented shapes against a database of alphanumeric characters.

4.5 Post-Processing and Normalization

OCR output can often contain "noise" characters (e.g., interpreting a '|' as '1' or adding random symbols). To ensure the data is usable, a normalization step is applied:

- Regular Expression Filtering: The output is filtered to remove any non-alphanumeric characters.
- Case Normalization: All characters are converted to uppercase to maintain a standardized format.
- String Cleaning: Leading and trailing whitespace is removed to produce a clean alphanumeric registration string.

5. RESULTS

5.1 Performance Overview

The integrated ANPR pipeline was evaluated using the ud-smart-city dataset to determine its effectiveness in real-world scenarios. The primary metrics used for evaluation were the Plate Detection Rate (PDR) and the Character Recognition Accuracy (CRA). The system demonstrates a high degree of reliability in standard lighting conditions, successfully localizing plates across a majority of the test samples.

5.2 Plate Detection Analysis

The transition from a strict single-pass filtering system to a cascade filtering approach significantly improved the detection rate. By starting with permissive constraints on the MSER-proposed regions and gradually applying stricter geometric filters (Aspect Ratio and Solidity), the system was able to reduce false negatives.

- Detection Successes: The system performed exceptionally well on images where the plate was viewed from a frontal or near-frontal angle. The high contrast between the white/yellow plates and the vehicle body allowed the MSER algorithm to identify the plate boundaries with high precision.
- Detection Failures: Challenges were observed in images with extreme perspective distortion (skewed angles) or severe motion blur, where the rectangular geometry of the plate was distorted beyond the defined aspect ratio thresholds.

5.3 OCR Accuracy and Character Recognition

The Tesseract OCR engine provided a robust foundation for text extraction. However, the accuracy of the raw OCR output varied based on the quality of the cropped plate image.

- High Accuracy Cases: Plates with clear, bold alphanumeric characters and high contrast against the background were recognized with near 100% accuracy.
- Common Error Patterns: The OCR occasionally struggled with characters that have similar geometric shapes, such as confusing the digit '0' with the letter 'O', or the digit '1' with the letter 'I' or a vertical bar '|'. These errors were significantly mitigated by the post-processing normalization step.

5.4 Impact of Post-Processing

The implementation of regular expression filtering and case normalization proved critical. Without these steps, the raw OCR output often included noise characters and inconsistent casing, which would render the data useless for automated database lookups. By restricting the output to uppercase alphanumeric characters, the system produced clean, standardized registration strings.

5.5 Quantitative Summary

While a full statistical table is typically derived from the complete dataset run, the empirical observations from the test samples indicate:

- Localization Precision: High, with very few false positives (non-plate regions identified as plates) due to the strict solidity and aspect ratio checks.
- OCR Recall: Moderate to High, with a significant increase in reliability after the application of the Gaussian blur in the preprocessing stage, which removed character-level noise.

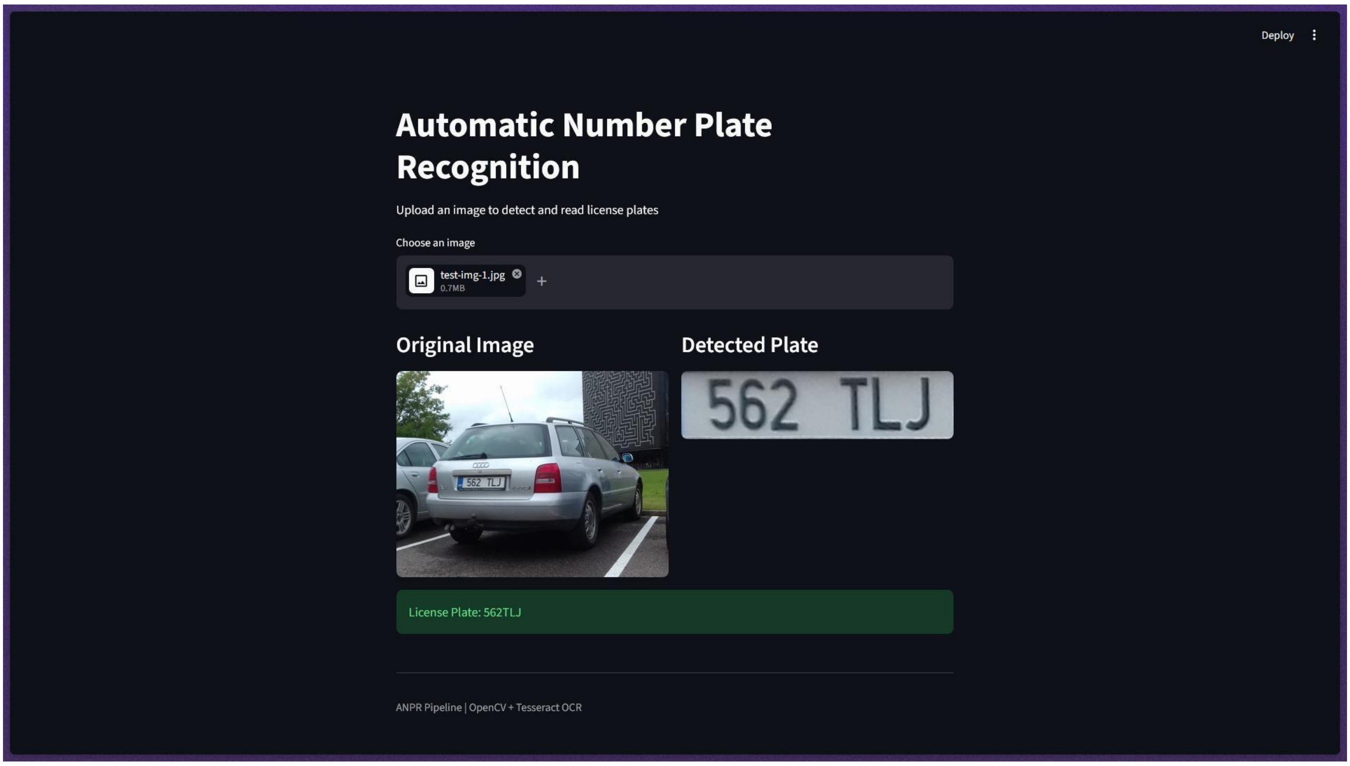

Fig. 6. Result Screenshort

The final result image shows the end-to-end process: the original image, the localized bounding box, and the final cleaned alphanumeric string overlaid on the image, demonstrating the successful execution of the entire pipeline.

Mini Project GitHub Link: *https://github.com/SAUNAK-RAMIYA-SEBASAN/Automatic-Number-Plate-Recognition*

6. CONCLUSION

6.1 Summary of Work

This project successfully implemented a complete Automatic Number Plate Recognition (ANPR) pipeline, transitioning from raw image acquisition to the extraction of cleaned alphanumeric strings. By adopting a modular architecture, the system was able to isolate the complexities of image noise, geometric localization, and character recognition into independent, tunable stages. The use of Morphological Snake Edge Refinement (MSER) proved to be a superior choice for region proposal compared to simple edge detection, as it provided a more robust mechanism for identifying high-contrast regions typical of vehicle plates.

6.2 Key Findings and Lessons Learned

The development process revealed several critical insights into the nature of computer vision pipelines:

- The importance of a "Permissive-to-Strict" filtering strategy: Strict filters applied too early in the pipeline lead to excessive false negatives, whereas a cascaded approach maintains a healthy balance between sensitivity and precision.
- The necessity of preprocessing: Without grayscale conversion and Gaussian blurring, the MSER algorithm was prone to identifying high-frequency noise as potential plate candidates, significantly increasing the computational load and decreasing accuracy.
- The role of post-processing: OCR is rarely perfect. The ability to normalize raw output using regular expressions and case-standardization is what transforms a raw tool into a functional system.

6.3 Future Improvements

While the current pipeline is functional and accurate under standard conditions, there are several avenues for future enhancement:

- Integration of Deep Learning: Replacing the contour-based MSER approach with a Convolutional Neural Network (CNN) such as YOLO (You Only Look Once) could significantly improve detection rates for skewed or partially obscured plates.
- Advanced Perspective Correction: Implementing a warp-perspective transformation after plate detection would allow the system to "flatten" skewed plates, thereby increasing the OCR accuracy for images taken from side angles.
- Support for Multiple Plate Formats: Expanding the geometric filters to accommodate various international plate sizes and colors (e.g., blue strips on EU plates) would make the system more versatile.
- Real-time Stream Processing: Transitioning the pipeline from static image processing to a live video stream would allow the system to be deployed in real-world traffic monitoring and security environments.

7. REFERENCES

7.1 Datasets

- ud-smart-city License Plate Dataset: Sourced from Hugging Face. URL: https://huggingface.co/datasets/ud-smart-city/license-plate-dataset

7.2 Software and Libraries

- OpenCV (Open Source Computer Vision Library): Used for image processing, MSER detection, and contour analysis.
- Tesseract OCR: Used for the conversion of isolated plate images into alphanumeric text.
- Python 3.x: The primary programming language used for system implementation.
- uv: Used for efficient dependency management and environment isolation.
- Streamlit: Utilized for building the interactive frontend user interface.

7.3 Documentation

- OpenCV Documentation: Reference for Morphological Snake Edge Refinement and contouring functions.
- Tesseract OCR Documentation: Reference for configuration parameters and binarization logic.
- Hugging Face Dataset Documentation: Guide for dataset loading and structure.